

WALLACE TREE MULTIPLIER

Vivado simulation

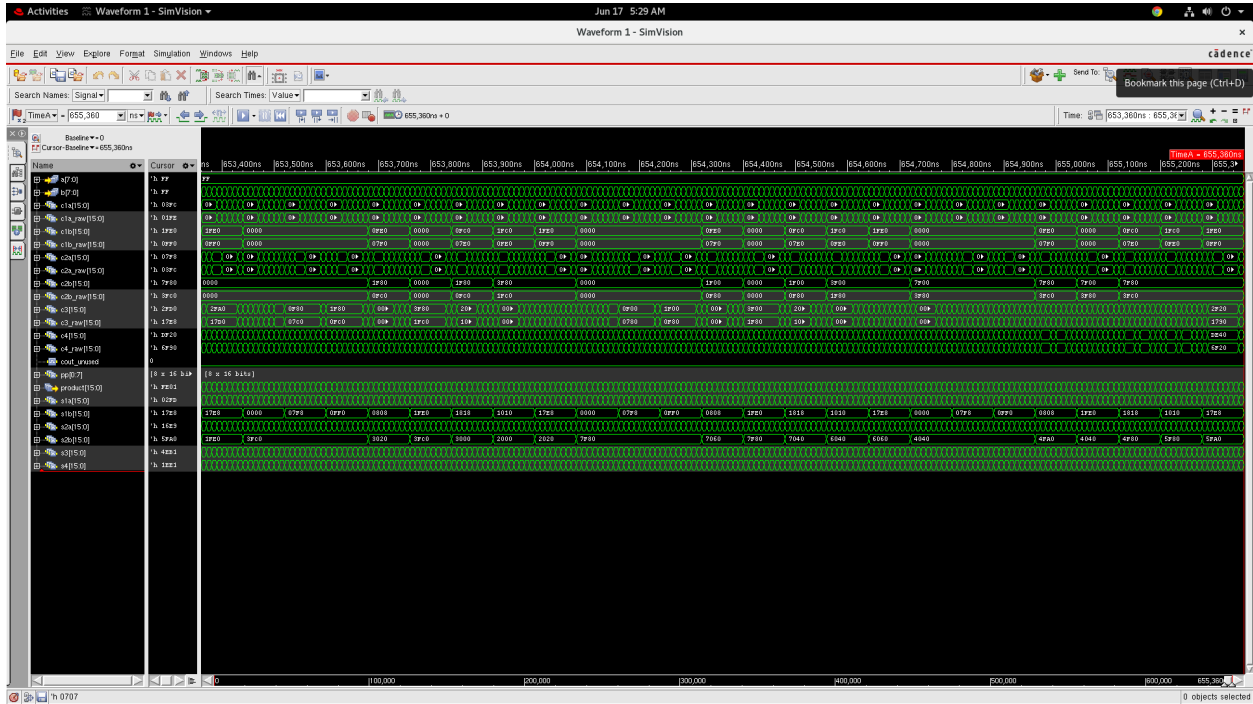


Figure 1. RTL Schematic After Elaboration

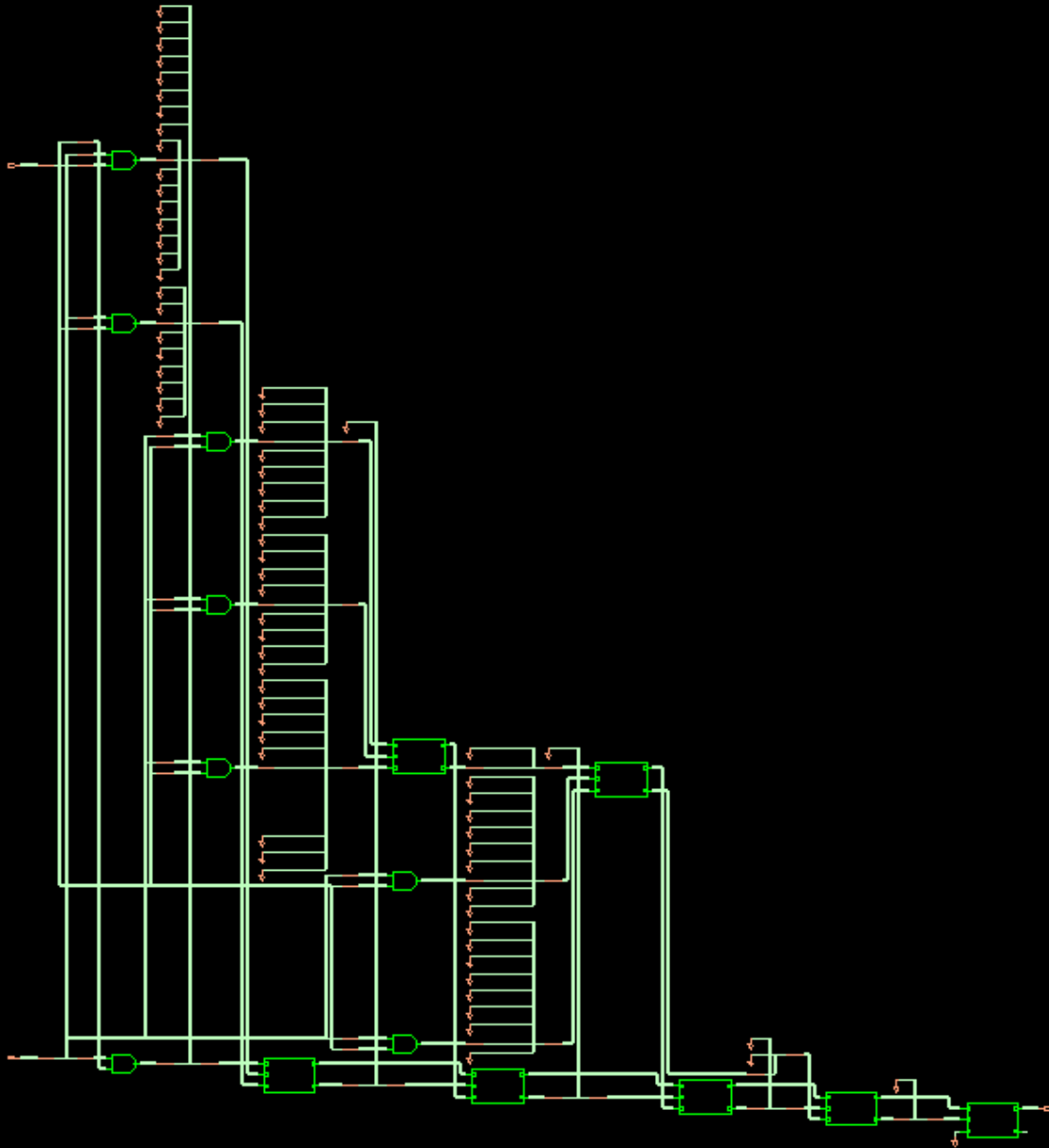


Figure 2. Generic Logic Synthesis

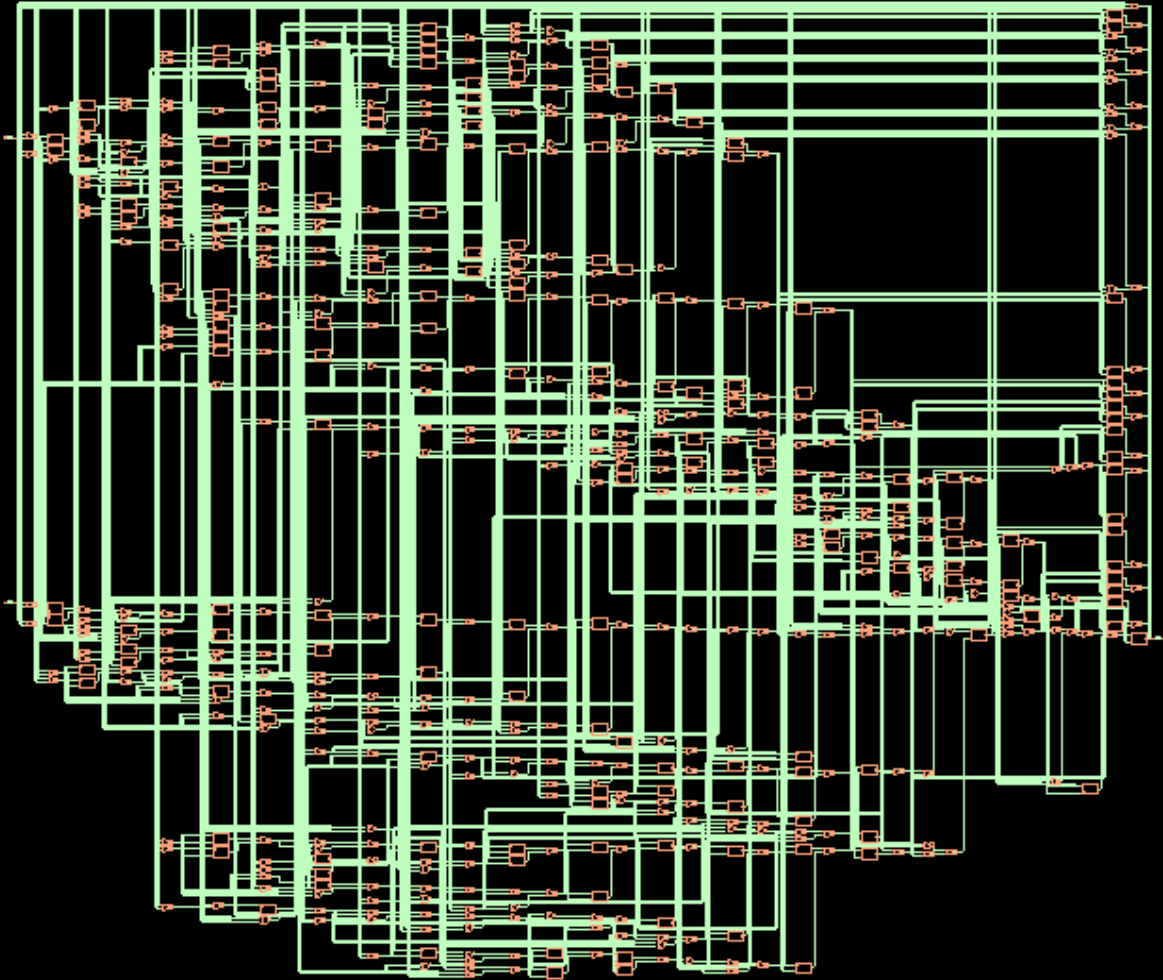


Figure 3. Technology Mapping to GPDK090 Standard Cells

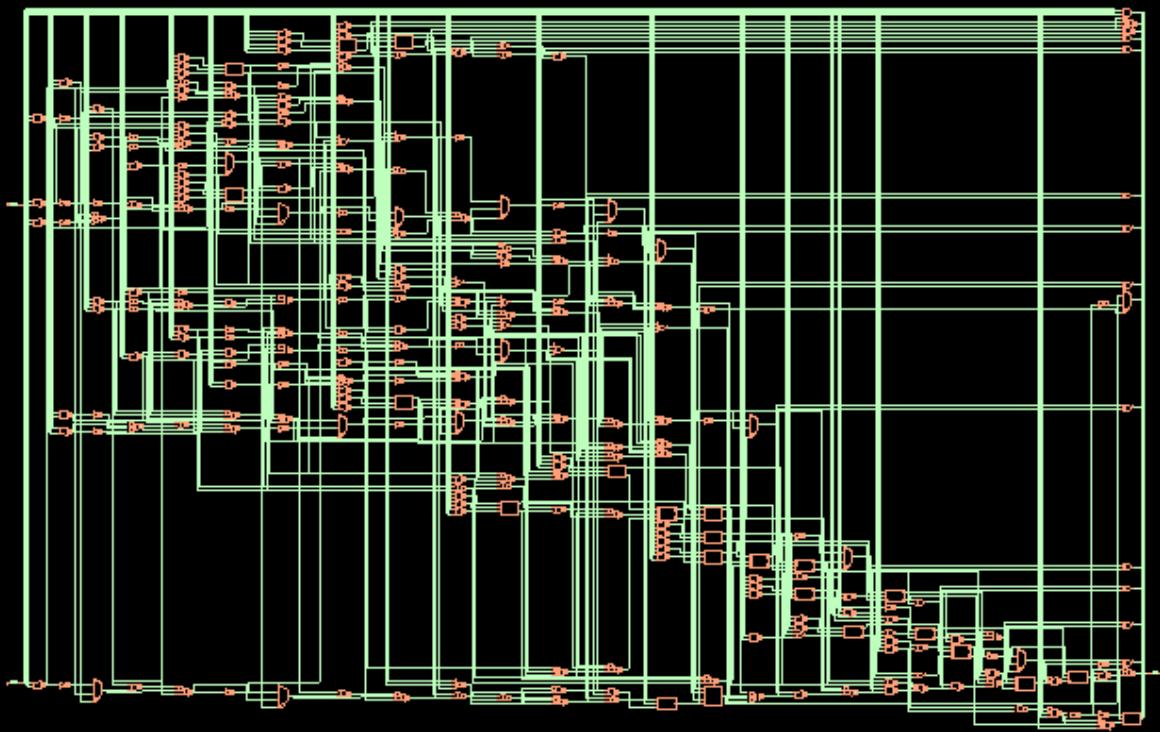
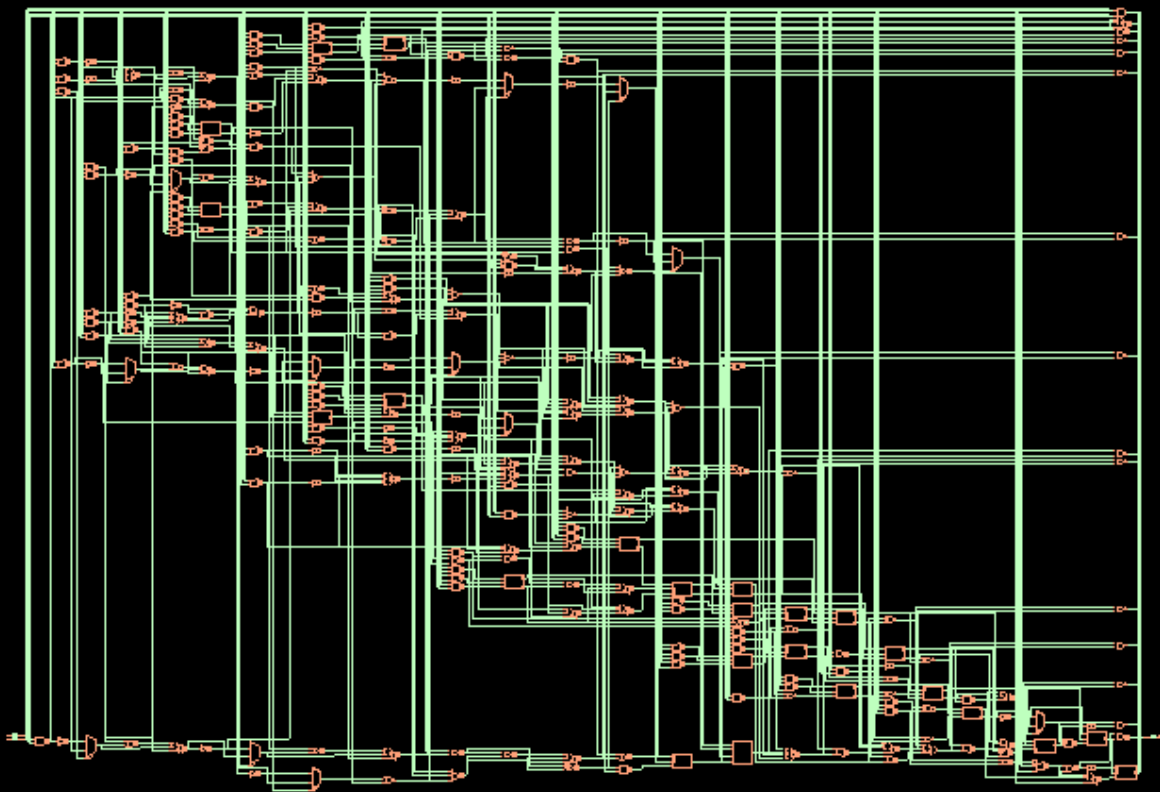


Figure 4. Post-Synthesis Logic Optimization



Genus Reports

File: /home/vlsi/4b4/bsp/simulation/area.rpt

Page 1 of 1

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=====
Generated by:      Genus(TM) Synthesis Solution 21.14-s082_1
Generated on:      Jun 17 2026 05:45:56 am
Module:            wallace_multiplier_8x8
Operating conditions: slow (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====
```

Instance	Module	Cell Count	Cell Area	Net Area	Total Area

wallace_multiplier_8x8		265	1720.434	0.000	1720.434 <none>
(D)					
(D) = wireload is default in technology library					

File: /home/vlsi/4b4/bsp/simulation/power.rpt

Page 1 of 1

Instance: /wallace_multiplier_8x8
Power Unit: W
PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	9.25579e-06	8.37387e-05	8.50318e-05	1.78026e-04	100.00%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%

Subtotal	9.25579e-06	8.37387e-05	8.50318e-05	1.78026e-04	100.00%
Percentage	5.20%	47.04%	47.76%	100.00%	100.00%


```

=====
Generated by:      Genus(TM) Synthesis Solution 21.14-s082_1
Generated on:      Jun 17 2026 05:46:31 am
Module:           wallace_multiplier_8x8
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

Timing

Clock Period

clk 5000.0

Cost Group	Critical Path Slack	TNS	Violating Paths
clk	0.8	0.0	0
default	No paths	0.0	

Total		0.0	0

Instance Count

Leaf Instance Count	265
Physical Instance count	0
Sequential Instance Count	0
Combinational Instance Count	265
Hierarchical Instance Count	0

Area

Cell Area	1720.434
Physical Cell Area	0.000
Total Cell Area (Cell+Physical)	1720.434
Net Area	0.000
Total Area (Cell+Physical+Net)	1720.434

Max Fanout	8 (b[7])
Min Fanout	1 (product[0])
Average Fanout	2.1
Terms to net ratio	2.9565
Terms to instance ratio	3.3358
Runtime	11.980045 seconds
Elapsed Runtime	261 seconds
Genus peak memory usage	1588.77
Innovus peak memory usage	no value
Hostname	vlsi-i5-166.cvr.ac.in